

Sebastiano Bacca

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PROFESSIONAL SYNOPSIS

As a doctoral researcher at the Global Climate Forum, I explore the socio-economic impacts of sea-level rise and develop innovative strategies for coastal adaptation. My research focuses on integrating nature-based solutions (NBS) with traditional infrastructure to address the twin challenges of climate resilience and ecological restoration. By employing economic modeling, econometrics, and environmental valuation techniques, I evaluate the costs and benefits of various adaptation scenarios, helping to identify pathways for sustainable and equitable solutions.

I hold a master's degree in environmental economics from Università degli Studi di Ferrara, where I earned recognition for academic excellence. My studies included an Erasmus program at Universidade do Minho, where I refined my expertise in economics and data analysis. Prior to my doctoral work, I contributed to the Núcleo de Investigação em Políticas Económicas e Empresariais (NIPE), focusing on the economic valuation of coastal and marine ecosystem services.

My interdisciplinary approach combines a robust background in environmental economics with practical insights into policy-relevant research. My goal is to empower coastal communities with actionable strategies for adaptation, ensuring resilience in the face of sea-level rise. Collaborating with scientists, policymakers and stakeholders, I aim to bridge the gap between academic research and real-world implementation, advancing sustainable development and climate resilience globally.

EXPERIENCE

Doctoral researcher | GCF – Global Climate Forum e.V., Germany July 2023 – Present

- Contributing with the GCF team to three European projects: DECIPHER, ACCREU, REST-COAST
- Working on global coastal wetlands valuation
- Implementing a global cost-benefit analysis of green and gray adaptation strategies
- Developing the economic and CBA module of the DIVA state-of-the-art model

Researcher | Universidade do Minho, NIPE, Portugal August 2022 – June 2023

- Part of the research team working on the CLICTOUR project
- Contributing to the implementation of a computable general equilibrium model to assess macroeconomic climate change impacts on Portuguese economy
- Environmental valuation of coastal ecosystem services in Esposende municipality, encompassing the marine protected area of the *Parque Natural Do Litoral Norte* situated in Northern Portugal

EDUCATION

Humboldt-Universität zu Berlin	PhD in Resource Economics Berlin, Germany	<i>July 2023 – Ongoing</i>
University of Ferrara	MSc in Environmental Economics Ferrara, Italy	<i>October 2020 – October 2022</i>
University of Minho	Erasmus+ program, MSc in Economics Braga, Portugal	<i>September 2021 – July 2022</i>
University of Bergamo	BSc in Economics Bergamo, Italy	<i>October 2016 – October 2020</i>

AWARDS AND CERTIFICATIONS

Excellence Award, Ferrara School of Economics and Management | University of Ferrara | Oct. 2022

ValueGaps Summer School, Economic Valuation of Natural Capital: From Theory to Practice | International Academy for Nature Conservation Isle of Vilm, Germany | 8-13 July 2024

SKILLS

Languages : Italian (Native), Bergamasque (native), English (C1), Portuguese (B2)

Professional : Cost-benefit analysis, Economic modeling and analysis, Econometric and statistical analysis, Research and technical writing

Personal : Problem-solving, Communication, Collaboration and teamwork, Critical thinking, Analytical and quantitative thinking

TECHNICAL SKILLS

Programming Languages : R, Python, Julia, MATLAB, GAMS

Software : Rstudio, Visual Studio Code, Gretl, QGIS, LaTeX (Overleaf), Zotero, Power BI, Tableau, any office software

Developer Platforms : GitHub, GitLab, Jupyter Notebook, Kaggle, Pluto Notebook, JuliaHub

PROJECTS

- DECIPHER** | *Decision-making framework and processes for holistic evaluation of environmental and climate policies* Ongoing
- European Commission, project number 101056898
 - GCF team - Economic valuation of coastal wetlands ecosystem services
- ACCREDU** | *Assessing Climate Change Risk in Europe* Ongoing
- European Commission, project number 101081358
 - GCF team - Cost and benefits of coastal adaptation strategies in the face of sea level rise using traditional and nature-based solutions
- REST-COAST** | *Large scale RESToration of COASTal ecosystems through rivers to sea connectivity* Ongoing
- European Commission, project number 101037097
 - GCF team - Coupling environmental valuation of coastal wetlands with the cost and benefit assessment of nature-based solutions implementation and restoration
- CLICTOUR** | *Climate change resilient tourism in protected areas of Northern Portugal* Concluded – June 2023
- Comissão de Coordenação e Desenvolvimento Regional do Norte (Portugal), NORTE-01-0145-FEDER-000079
 - Economic valuation of coastal ecosystem services in Northern Portugal marine protected area

PUBLICATIONS

- Sousa, R., Vieira, A., Silva, L., Ribeiro, S., Melo, A. D., Bento-Gonçalves, A., Viana, A., Gião, C. G., Machado, C., Carballo-Cruz, F., Diniz, G., Pereira, H., Freitas, I., Cerejeira, J., Novais, J., Ribeiro, J. C., Veiga, L. G., Aguiar-Conraria, L., Ochoa, L. E. E., ... Marques, T. (2023). *CLICTOUR - Roteiro para o turismo sustentável nas áreas protegidas do norte de Portugal* (Rep.) (Accepted: 2023-10-19T11:37:35Z). <https://repositorium.sdum.uminho.pt/handle/1822/86994>
- Palazzo, A., Nakhavali, A., Gerber, G., Arbelaez Gaviria, J., Jung, M., Drews, M., Meuriot, O. G. M., Martin, J. S., Bacca, S., Hinkel, J., & Lincke, D. (2025). ACCREDU Report: Impacts on ecosystems & biodiversity [Publisher: ACCREDU (Assessing Climate Change Risk in Europe)]. <https://www.accreu.eu/news/accreu-report-climate-change-and-land-use-impacts-on-biodiversity-ecosystems-and-their-economic-value/>
- Haas, J., Lincke, D., Bacca, S., Honsel, L., Hinkel, J., Binet, S., van Ginkel, K., Jansson, P., Tesselaar, M., Ingels, M., Botzen, W., & Brusselsaers, J. (2025, February). *Impacts on infrastructure and built environment*

May 17, 2026
Sebastiano Bacca